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Checks-in, and Balances: a Matter of Time

Overview

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This is the model overview required as a total rundown, written in line with the ground rules: it uses the variable list (`Variable_Names.md`) and definitions list (`Definitions.md`), refers to assumptions, equations, constraints and tables by their captioned numbers (maintained in `Methodology.md`), and is authored and dated per Ground Rule 4.

I Purpose and scope

The model forecasts the build-up of departing passengers at check-in so that wait times can be estimated and tested against service targets. The scope is a single operating day, Day 7 (21 June 2026), of international departures from Delhi (DEL) Terminal 3. After cleaning, the schedule contains 164 flights and a maximal 39,610 seats (N, Equation (1)).

The model is built in layers: a fixed schedule foundation, a seat universe with a load-factor-driven existence overlay, a set of entry-time models, a check-in desk and service representation, and a queue and wait computation assessed against the OMDA benchmark.

II Schedule foundation (Layer A)

The foundation schedule `asimplified.csv` holds one row per flight, each described by the core author-defined variables: `flight`, `n`, `route`, `dest`, `cap`, and `dep`. The serial `n` runs 1 to `n_last` (= 164) in departure order. One flight, B4 443 (Beond, DEL-MLE), is omitted because the source gives it neither a Ground Handler nor a Capacity (Methodology, Data Note); the original is preserved as `asimplified_with_B4_443_raw.csv`. AIESL is treated as a Ground Handler independent of AISATS.

III Seats and existence (Layer B)

Step 1 enumerates the universe: one occupiable seat per physical seat, assigned many-to-one to its flight, indexed `seat` = 1 to N in departure order (Assumption 1, Equation (1)). The result is the seat-level master `seats.csv` (Table 1), with 39,610 rows (Table 2).

Real demand is then thinned by a load factor. Under Assumption 2 and Equation (2), each flight realises $\text{round}(\text{LF} \times \text{cap})$ passengers (`pax`), its lowest-indexed seats, and the rest are marked `exists` = 0. Because LF is a free parameter (for example 0.90), the realised passenger count is deliberately variable across scenarios; unoccupied seats never enter the system.

IV Entry-time models (Layer C)

Every occupied seat (a `pax`) is assigned an entry time, the moment they enter the check-in area, drawn from a model `m`, which is a distribution over the flight's check-in window [`dep` minus 4h, `dep` minus 1h] (Equation (3)). Several models can coexist (M1, M2, ...), each distributing entry times differently; unoccupied seats have no entry time. This layer is where the bulk of modelling judgement lives, and is the natural place to use the existing T3 check-in work (`pre0618/T3 Checkin Model, Pax Arrival Times.xlsx`).

V Check-in desks and service (Layer D)

On the supply side, each airline `a` opens `D(a, t)` check-in desks at time `t`, and `D` may change through the day. Desks may be open only within [`dep` minus 4h, `dep` minus 1h] (Constraint (1)), which the data confirms equals [`checkinopen`, `checkinclose`] for every flight. Each desk `k` is a point with processing speed `tau` (seconds per passenger), or rate $\mu = 1 / \text{tau}$ (Equation

(4)). For now all desks share the same speed and capacity (Assumption 3); this homogeneity will later be made idiosyncratic by airline and by GHA.

VI Eligibility, queueing and wait (Layer E)

Not every passenger may use every desk. The eligible set $E(p)$ is governed by the passenger attributes `loyaltystatus` (Table 3), `class` (Table 4), and `fastforward`. At a given desk the wait is the work still ahead: passengers ahead times service time, less the time already spent on the passenger in service (Equation (5)). A rational passenger joins the eligible desk with the least wait (Assumption 4, Equation (6)). The resulting waits are judged against the OMDA Target (Table 5): at most 5 minutes for business class, at most 20 minutes for economy.

VII Variables, assumptions and equations

The full variable registry is in `Variable_Names.md` and all terms in `Definitions.md`. The model rests on four assumptions (1 one passenger per seat, 2 the existence tail rule, 3 homogeneous service, 4 rational counter choice) and one operating constraint (1) on the desk window. Its quantitative spine is Equations (1) to (6): universe size, existence, entry-time distribution, service rate, wait at a desk, and chosen wait. All are catalogued in `Methodology.md`.

VIII Artifacts

`asimplified.csv` (cleaned foundation schedule, 164 flights); `seats.csv` (seat-level master, 39,610 rows, Table 1); `seats.xlsx` (seat-level master with entry-time model columns M1 to M13); `build_passengers.py` (reproducible Step-1 build); `asimplified_with_B4_443_raw.csv` (provenance); `flight_table.Rmd` (rendered schedule view); `curves.Rmd` (rendered `curves.pdf`: M1 entry-time distribution per flight, one per page); and the governance set `00_Ground_Rules.md`, `Variable_Names.md`, `Definitions.md`, `Methodology.md`, and this `Model_Overview.md`.

IX Status

Built and verified: Layers A and B (schedule foundation and seat universe, Step 1); Layer C entry-time models (Step 2 build note: 104 models, N1 to SR39, instantiated in `seats.xlsx` and verified in `verification_report.csv`); and, in `assumed statics/`, the Layer D and E wait engine and counter allocation (Equations (7) to (16)): FIFO waits, per-minute staffing profiles, two service regimes (180 s and 200 s per passenger), two targets (20 and 40 minutes), three pooling granularities, and the comparison against the physical supply of 168 counters (headline: at 180 s per passenger the 20-minute OMDA target fits within 168 counters under every pooling; arrival shape, not counter count, governs feasibility).

Still open, as listed in `Methodology.md` and `assumed statics/Methodology_Counter_Allocation.md`: load-factor scenarios below 1, desk eligibility by class and loyalty (the mapping $E(p)$), island and 14-per-island supply constraints, and the tighter time-varying optimum.